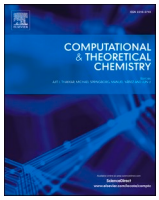
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Computational insights into the photophysical processes of an acylhydrazone fluorescent probe based on conical intersection and sensing mechanism for Al^{3+} detection

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A B S T R A C T

Al^{3+} detection is crucial for food safety, environmental monitoring, and biological assays. This work investigated the fluorescent properties of the probe VN62 and its sensing mechanism for the detection of Al^{3+} ion in detail. The ground- and excited-state properties such as isomerization geometries and absorption spectra, as well as the photophysical processes of transitions in PES have been studied. There are few fluorescent S_1 states (PT*, TICT) of the VN62 molecule, and the conical intersections (CIs) have key importance in the fluorescent process. By comparing the radiative rate (k_r) and the rate of transition to CI, we deduce that these fluorescent states are lack of stability for light emission. Therefore, the fluorescence of probe alone can be quenched by CI pathways. After coordination with Al^{3+}, the CI pathways will be suppressed, resulting in enhanced fluorescence. This sensing mechanism is different from the previously proposed mechanisms, such as the dark state of TICT or ESIPT. We anticipate that our research would shed some light on the design of turn-on sensors.

1. Introduction

Aluminum is the most abundant metal element in earth's crust and is widely used as food additives, water purification agents, and medicines. An excessive intake of aluminum can bring damages to human brain and may cause illnesses such us Alzheimer's disease, Parkinson's disease, dialysis encephalopathy, and so on [1–3]. According to the world health organization (WHO), the maximum allowed concentration of Al^{3+} in drinking water is 7.4 μmol/L [4]. Therefore, sensitive detection of Al^{3+} is of great importance in both food safety management, environmental monitoring and biological assays [5].

In recent years, the fluorescence analysis method has garnered significant attention due to its distinctive advantages, including high sensitivity, excellent reproducibility, minimal sample consumption, user-friendly operation, and reduced time requirements. A series of fluorescence-based sensors have been developed for detection of Al^{3+} ions, [6–10] and the Schiff bases with arylhydrazone, acylhydrazide, or acylhydrazone functional group, [11] which are good ligands to coordinate with Al^{3+}, [12] have been widely used. Unfortunately, it is noteworthy that most of these probes are limited to organic solvents, thereby restricting their application in aqueous environments and biological samples.

Based on previous studies, the sensing mechanism of fluorescent probe like acylhydrazone molecule has been carefully investigated. Several important photophysical processes has been discussed in the excited-state dynamics: such as photo-induced electron transfer (PET), [13–15] excited state induced proton transfer (ESIPT), [16–20] twisted intramolecular charge transfer state (TICT), [21–23] and conical intersections (CIs) [24–26]. In general, the fluorescence of the sensor itself was very weak, and it was explained by the PET process in early experimental works. Treto-Suárez and co-workers calculated PET rates and emission rates of the probe and the corresponding metal coordinates, and they explained the selectivity of Al^{3+} detection by the PET mechanism [27]. In recent works, [28–30] ESIPT and TICT states have been considered in the excited-state dynamics, and it was deduced that the fluorescence of the probe is quenched by the TICT dark state. Back in 2016, Budzak and Jacquemin have investigated the CI structure in an acylhydrazone probe, [31] and they suggest that the constraining of CI is responsible for the strong fluorescence turn-on after the formation of complexes between the probe and the Al^{3+} ion.

Lately, we have synthesized a new fluorescent probe VN62 [32] for selectively detection of Al^{3+} in aqueous medium. The structure of VN62 is shown in Fig. 1. The functional group acylhydrazone (in red circle), together with the hydroxyl in vanillin, are the key structure for Al^{3+} detection. The probe molecule has good structural stability and water solubility benefit by the hydroxyl groups, and thus can be applied for

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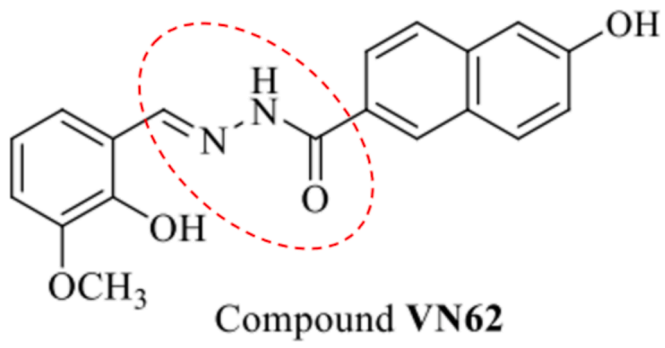


Fig. 1. Structure of fluorescent probe VN62.

plant and live-cell imaging. Besides, the probe exhibits good performance such as high specificity, a fast response time, wide pH response range, good anti-interference, and high sensitivity, with a quite low detection limit of 39.6 nmol/L.

To gain a deeper understanding of the electronic structure and structure–activity relationships of the acylhydrazone fluorescent probes, the photophysical processes of VN62 and the fluorescent sensing mechanism for Al^{3+} detection will be theoretically studied in this paper. We mainly focus on the ground- and excited-state properties of the possible conformations of VN62 and the spectral properties of the Al^{3+}-VN62 complex. We verify the existence and key importance of the CI structure, PT and TICT states in the first excited singlet state (S_1) and the ground state (S_0) of the probe. By calculating the radiative rates and the transition rates in S_1, we will analyze the possible photophysical processes of fluorescence quenching, and reveal the sensing mechanism of VN62 detecting Al^{3+}.

The paper is organized as follows. In section 2, we briefly outline a general description of the computational methods used in this paper. Section 3 illustrates the results and discussions, including detailed calculations on the probe alone and the Al^{3+}-VN62 complex. The ground state conformations of VN62 molecules are searched to find the stable structure of the probe. Absorption spectra and excitation electron-hole distribution are calculated to illustrate the excited-state properties. Then, the fluorescent S_1 states of VN62 are carefully investigated to reveal the quenching mechanism. Finally, the geometry and spectra of Al^{3+}-VN62 complex are given to complete the sensing mechanism. In the end, our conclusions are given in section 4.

2. Computational detail

Density functional theory (DFT) and time-dependent DFT (TDDFT) are used to obtain the geometries and energies of the probe in the ground state and electronic excited state. Considering the weak interaction of hydron bond and the conjugated aromatic structure in our systems, as well as the charge transfer state in excited states, conventional functionals like B3LYP may not be able to correctly describe the electronic structure. Hence, dispersion correction is considered using the range-

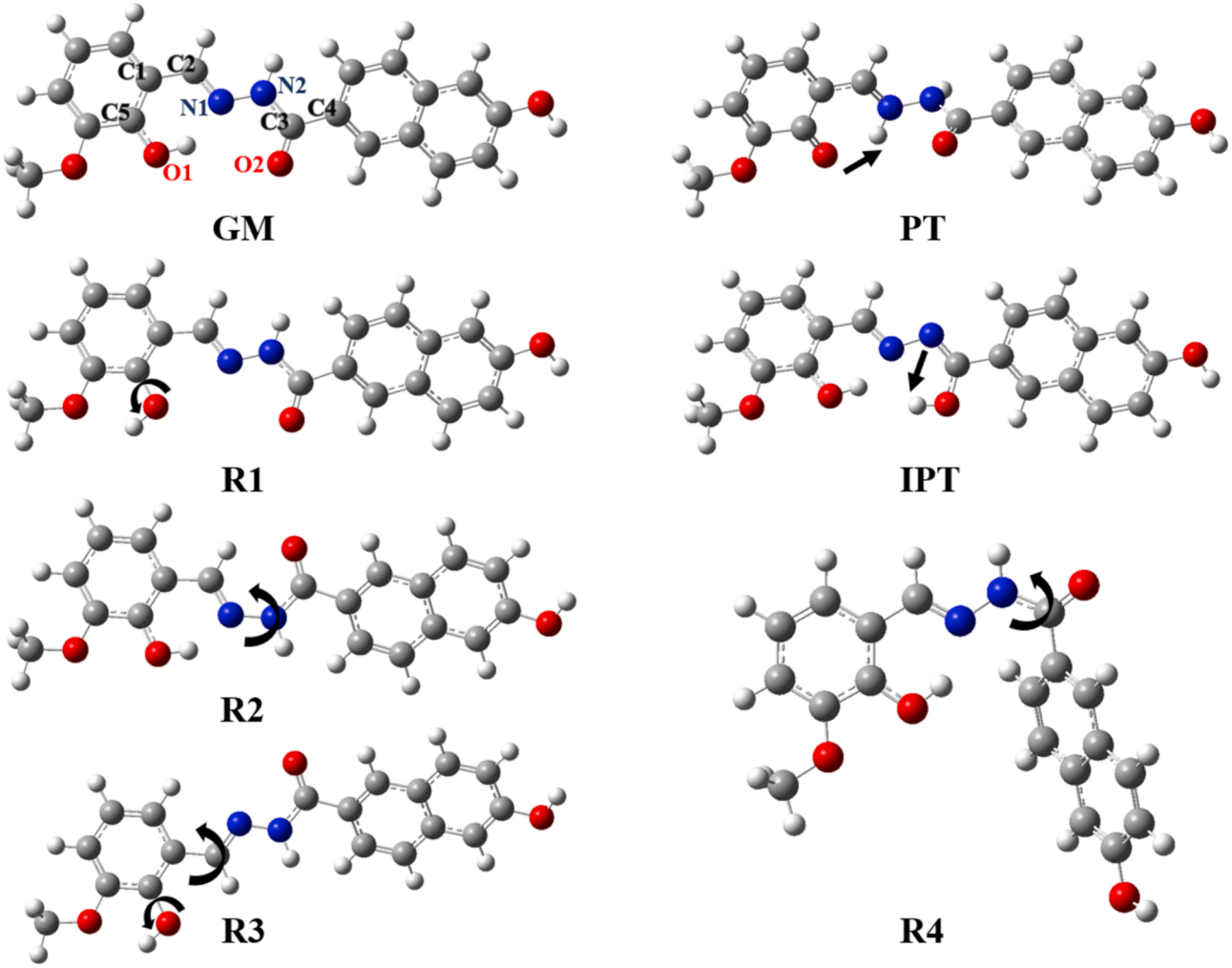


Fig. 2. Global minimum and local minimums of VN62 structures. The C, N, O, H atom are represented by grey, blue, red and white, respectively.

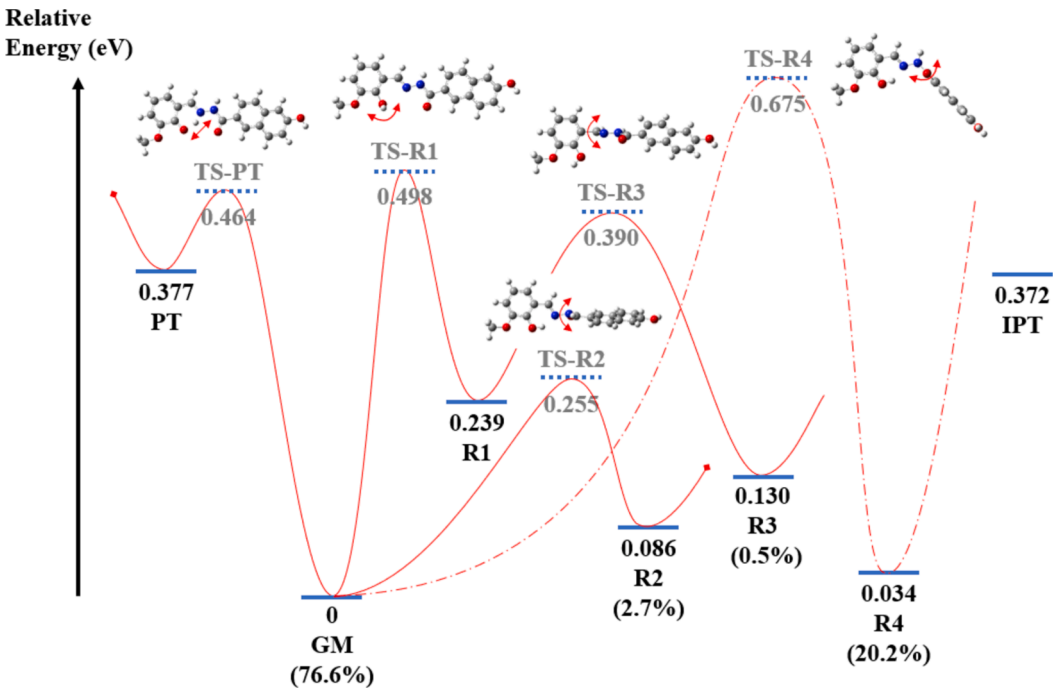


Fig. 3. Ground-state energy diagram of the local minimums and transitions between them. The local minimums and transition states are distinguished by black and grey color.

separated functional ωB97XD [33] and the basis set 6–311 g(d, p) is applied throughout the calculations. The solvent effect is considered using the CPCM model [34] with water as the solvent. A broaden factor of 0.15 eV is used to obtain the spectral line shape. All the ground-state calculations and the absorption spectra are performed within the Gaussian 16 software package [35].

When we consider the CI structures on the potential energy surfaces (PES) of the ground state and excited states, the single-referenced DFT method will be hard to describe the multi-PESs character. For the calculations of the optimization of S_1 excited states and CI structures, all the jobs are performed with both regular TDDFT method and spin-flip (SF) TDDFT method [36] in ORCA software packages, [37] and the similar functional ωB97X-D3 are used [38]. The basis set 6–311 g(d, p) and the CPCM model are used, same as the previous Gaussian jobs. All the structures in PES are constructed in the help of package Gentor [39]. Electrostatic potential analysis, and electron – hole distribution analysis are performed within tool package Multiwfn [40]. The software VESTA is also used to visualize the molecular structures and orbitals [41].

3. Results and discussion

3.1. Geometries and PES in the ground state

The optimized structure of VN62 is shown in Fig. 2 (upper-left panel). This organic molecule is very flexible, and molecular rotation can form many conformational isomers. The rotation can happen along bonds between C1, C2, N1, N2, C3, C4 atoms, especially single bond like N1—N2. Also, the presence of the hydroxyl group −O1H may induce proton-transfer process with the adjacent N1 atom. Thus, the ground state of VN62 should have several local minimums, making the ground-state PES quite complicated. Herein, the ground-state PES is explored to locate the global minimum (GM) by scanning the related dihedrals and the H atom position between O1 and N1.

As plotted in Fig. 2, seven lowest local minimums are observed and named as GM, R1, R2, R3, R4, PT and IPT. GM structure has the lowest energy, due to the formation of hydrogen bond O1H…N1. The R1 structure is constructed by rotating the C1C5-O1H dihedral angle, till the −O1H group facing to the other side (180°). The R2 structure is constructed by rotating the C2N1—N2C3 dihedral angle, with the O2 atom approaching to the C2H side, eventually forming the second hydrogen bond. The R3 structure is constructed by rotating both the C1C5-O1H and C5C1—C2N1 dihedrals to 180°. All the above molecular rotations are along single bond, so the energy barrier of isomerization should be relatively low. In the GM structure, the bond length of C2—N1 is 1.28 Å, which is a typical C═N double bond in hydrazone. And the bond length of N2—C3 is 1.37 Å, a regular C—N bond in amide, which is between single bond and double bond. It is hard to rotate molecule along these two bonds, and we will thoroughly study the drastic changes of the molecular PES when we do the rotations in the following content. It is also known that the C═N double bond can undergo E/Z isomerization, [42] and the structure R4 will be constructed by rotating N1N2—C3C4 dihedral angle to 180°.

On the other hand, isomerization can also be caused by proton transfer (PT). The PT structure will be constructed by moving the hydrogen from O1 atom to N1 atom, which turns the molecule to an imine structure. In the meantime, the molecule will be rotated along N1—N2 bond to minimize the energy of the system, and the C2N1—N2C3 dihedral angle in PT will be about 101°. The conformation of PT from O1 to O2 has a very high relative energy, so it is not shown here. At last, indirect proton transfer (IPT) process can also happen from N2 to O2 atom. The formation of IPT structure may go through R4 structure as an intermediate.

The relative energies of the seven local minimums (GM's energy is set to be 0) and the transition states (TS) of ground-state isomerization reactions are calculated and listed in Fig. 3. In this diagram, solid lines represent the local minimums, and dashed lines represent transition states. The structures of TS are shown above the corresponding states. The Boltzmann distribution is also calculated and listed in parentheses under the certain conformations. The following equation is used: [28]

$$P_i = \frac{e^{-\frac{\Delta E_i}{RT}}}{\sum_j e^{-\frac{\Delta E_j}{RT}}}$$

P_i is the probability of structure i comparing to all the possible

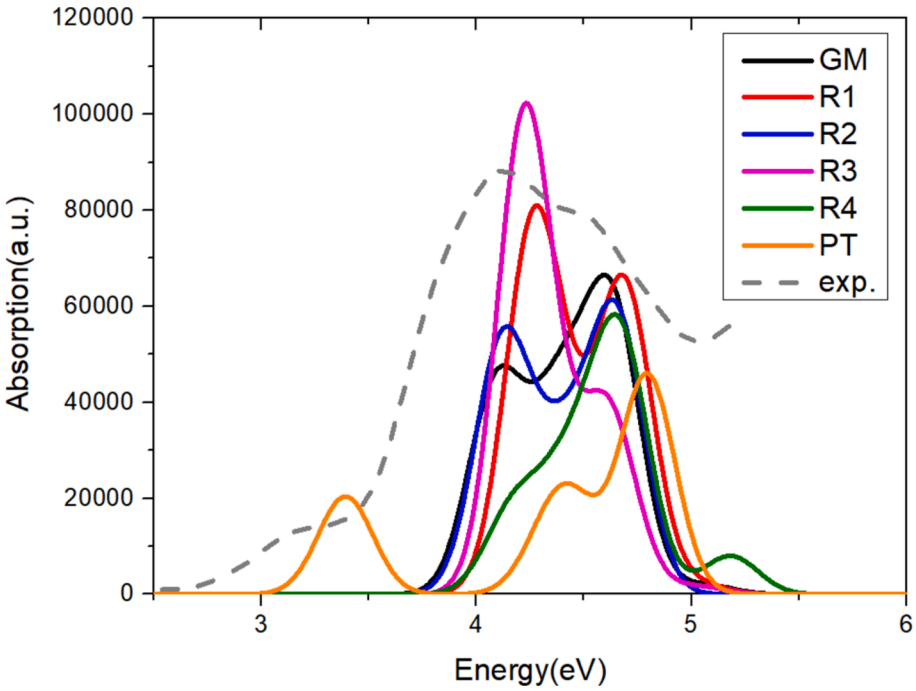


Fig. 4. Absorption spectra of GM, R1, R2, R3 R4 and PT in water (broadening factor 0.15 eV) and the experimental spectrum of VN62 solution.

conformations, and ΔE_i is the energy difference between structure *i* and the most stable structure. The temperature is set as 300 K. The ground state structure of VN62 in thermodynamic equilibrium is mainly GM (76.6 %). The R4 structure has a proportion of 20.4 %, and The R2 structure only has 2.7 %. Other structures like R1, R3, PT and IPT have very small proportions.

Besides the thermodynamic effect, the kinetic effect of isomerization should be considered. As we expected before, the energy barrier of TS-R4 has the highest energy of 0.675 eV, due to the E/Z isomerization undergoes the rotation along N2—C3 amide bond. And the energy barrier of TS-R2 and TS-R3 is 0.255 eV and 0.151 eV, respectively. Rotations along single bond (N1—N2, C1—C2) are much easier. The barrier of TS-R1 is 0.498 eV, higher than single bond rotation, because of the hyperconjugation effect of the phenolic hydroxyl −O1H. Lastly, the barrier of TS-PT is 0.464 eV (10.7 kcal/mol), which can be easily overcome in ground state reaction. This energy barrier can also be calculated by scanning N1—H bond. It should be noted that GM and R2 can form the same PT structure through similar process (see in Figure S1).

In summary, the GM structure is the most stable structure of VN62, but the energy barriers of transitions between other conformations are not very high (less than 20 kcal/mol). So the conformations caused by molecular rotation and the isomerization may have influence in the fluorescent processes of the probe.

3.2. *Absorption spectra*

The absorption spectra of above structures are calculated using TDDFT method, and shown in Fig. 4. The first five low-level singlet excited states are considered, and the information of these excitations can be found in Table S1. The main absorption peaks of all the conformations of VN62 (except for PT) are distributed in 4–5 eV, which matches well with the experimental data. The line spectra of GM, R1, R2, R3 and R4 without broadening are shown in Figure S2, and they can be

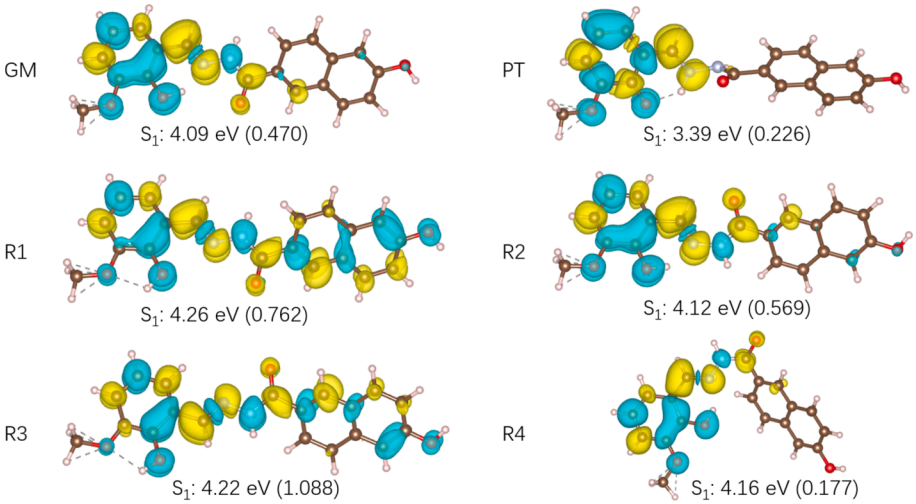


Fig. 5. Electron − hole distributions of S_1 states of the ground-state conformations (blue for electron, yellow for hole. Isovalue = 0.001).

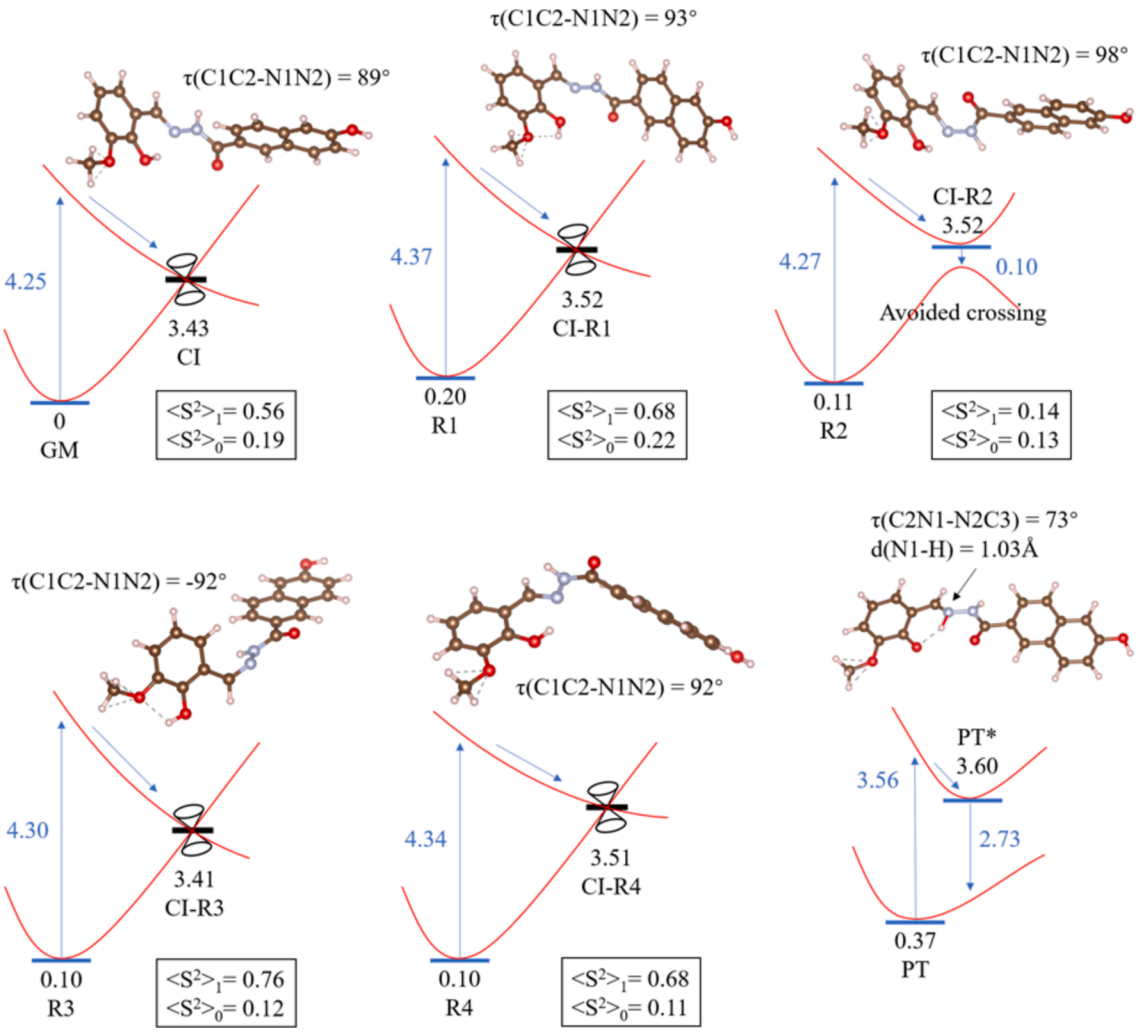


Fig. 6. Energy diagram of VN62 (the S_1/S_0 energies are relative to ground-state GM structure). The excitation energies ΔE_{01}, ΔE_{10} are shown in blue (all energies are in eV), and the structures of CI or PT* are shown above the diagrams.

divided in two bands. The low energy band is contributed by S_1 state (marked as dark red), and the high energy band is contributed by S_4 state (marked as light blue). The oscillator strength of S_1 is smaller than S_4 for GM, R2 and R4, while the situation for R1 and R3 is opposite, showing different line shapes in the absorption spectra. According to the line shape of the experimental spectrum, we assume that the conformation of R1 or R3 should have a proportion in VN62. On the other hand, there is also a shoulder around 3.3 eV in the experimental spectrum. This can be explained by the contributions of PT structure or the anion form of the probe molecule.

The contributions of molecular orbital (MO) transitions for all excitations have been analyzed, and the frontier MOs of all conformations are shown in Figure S3. For excitations of GM, R1, R2, R3 and R4, the MO transitions are similar: S_1 state is mainly contributed by HOMO to LUMO transition, S_4 state is mainly contributed by HOMO-2 to LUMO transition, S_5 state is mainly contributed by HOMO-4 to LUMO transition. However, these excitations cannot be described by single transition of MO pair. So, we further use the electron-hole analysis to characterize the excitations. The electron − hole distributions of S_1 states are plotted in Fig. 5. The corresponding excitation energy and oscillator strength (in parenthesis) are also listed. We can divide the molecule into vanillin part and naphthol part, and according to the picture, the S_1 state can be considered as local excitation of π-π* transition. For R1 and R3, the electron − hole distribution spreads around all parts of the molecule. For GM, R2 and R4, electron and hole are mainly distributed in vanillin part and some in the connecting part. For PT, the electron − hole distribution only covers the vanillin part, where the vanillin plane is perpendicular to the naphthol plane. The PT structure has the lowest excitation energy (3.39 eV), and this excitation of S_1 state can also be described by the transition from HOMO to LUMO, which shows the same character of local excitation.

3.3. *Optimization of S_1 states and the S_0/S_1 CI structures*

In general, the fluorescence is emitted from S_1 state based on Kasha's rule [43]. The fluorescent property of probe is determined by the S_1 state. When we optimize the S_1 states of above structures of VN62, only PT* has an ordinary S_1 local minimum, and GM, R1, R2, R3, and R4 structures undergo the CI structures, where excitation energy between S_0 and S_1 are nearly 0 [24]. The system can be de-excited efficiently by these non-radiative CI pathways. The energy diagram is shown in Fig. 6. During the SF-TDDFT calculations, the spin expectation vales are checked to be smaller than 1. Considering the accuracy of ωB97X-D3 is improved compared to ωB97XD in terms of weak interaction, the ground-state energy and S_1 excitation energy are slightly different with aforementioned (ωB97XD).

The CI structures are named as CI, CI-R1, CI-R2, CI-R3 and CI-R4, related to the ground-state local minimums. They can be achieved by molecular rotation along C2—N1 bond, and eventually the dihedral angle of C1C2—N1N2 is around 90°. Particularly in the CI-R2 structure,

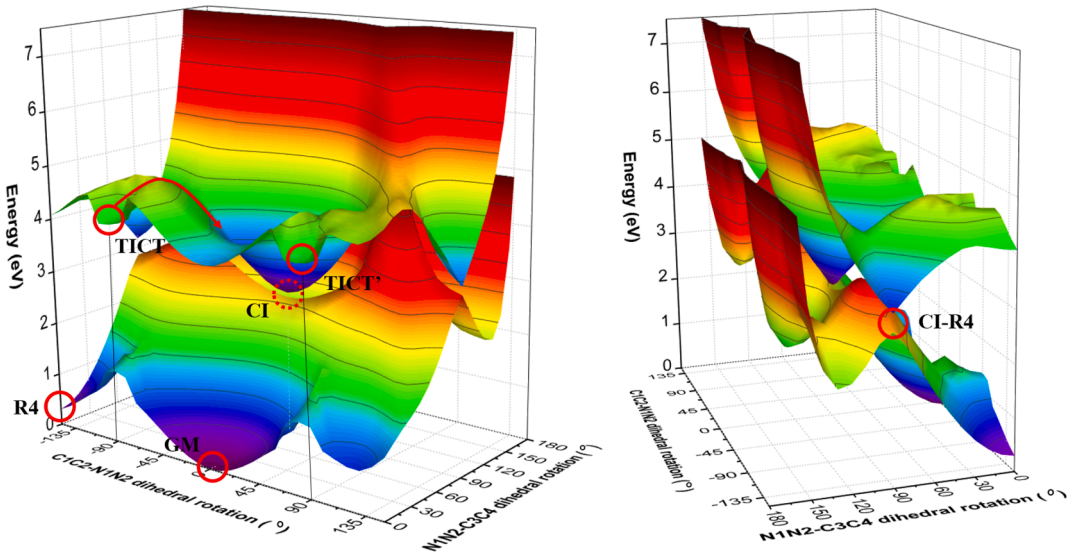


Fig. 7. PES around TICT state and CI, colored by the relative energies (0–5 eV for S_0 surface, 2.5–7.5 eV for S_1 surface). Front view and back view.

we find an avoided-crossing feature in PES, where the S_0/S_1 excitation energy is only 0.1 eV. The formation of CI structure requires the C═N double bond in hydrazone, and the O1H…N hydrogen bond cannot block the CI pathway.

The PT* structure is like PT structure in ground state, with a smaller C2N1—N2C3 dihedral of 73°. As the N1—H bond is forming, the C2═N1 double bond is weaken. Therefore, the formation of CI structure will be restricted. On the other hand, the energy gap between PT* state and ground state is 2.73 eV, and the transition has a non-zero oscillator strength (f = 0.243), which means that it still has a potential of fluorescence.

3.4. Possible photophysical processes of fluorescence quenching

To further reveal the influence of molecular geometry in the isomerization, excitation and de-excitation processes of VN62, and to find the relations between the S_0/S_1 states, the PES was thus constructed by scan the related dihedrals and bond lengths in the molecule. Firstly, we do rigid scans of the C1C2—N1N2, C2N1—N2C3 and N1N2—C3C4 dihedrals starting from GM structure, the variations of S_0/S_1 energies are plotted in Figure S4. We can easily locate the GM, R2, R4 in the S_0 curves (not the real minimums), and the CI feature is also observed when the C2N1—N2C3 dihedral is 90°. Besides, a new TICT state is located in S_1 curve when the N1N2—C3C4 dihedral is 90°. The TICT state and its image structure TICT' states are optimized, followed by the excitation analysis. The electron − hole distribution confirms the CT character of electron transfer from naphthol part to vanillin part (see in Figure S5). The TICT and TICT' states have a small but non-zero oscillator strength (f = 0.034, 0.044). So, the fluorescence quenching cannot be explained by formation of TICT dark states.

Now we scan both the C1C2—N1N2 and the N1N2—C3C4 dihedrals starting from CI structure, and the PES, which is consisted of the variations of S_0/S_1 energies, are plotted in Fig. 7. In the picture, the conical intersection is shown more intuitively: the surfaces of S_0 and S_1 are touching in CI position. Also, all the GM, R4, TICT and TICT' states can be located in their corresponding positions. The PES is also calculated by regular TDDFT, and plotted in Figure S6. The curves of the PES are similar, only that the energies are slightly different from the SF-TDDFT results. Especially, there is an obvious gap between S_0 and S_1 states in CI point.

Based on the PES of the system, we can find a possible pathway between the TICT state and the CI in S_1 surface. Starting from the TICT state, the system can go uphill to overcome a little barrier and then go downhill to reach the lower CI position (red arrow in Fig. 7). The related transition state (TS*-TICT) is calculated and the energy diagram is plotted in Figure S8. In this way, the system will go through a non-radiative pathway, and the possible fluorescent state will be quenched.

When we scan the C1C2—N1N2 dihedral and the N1—H bond length

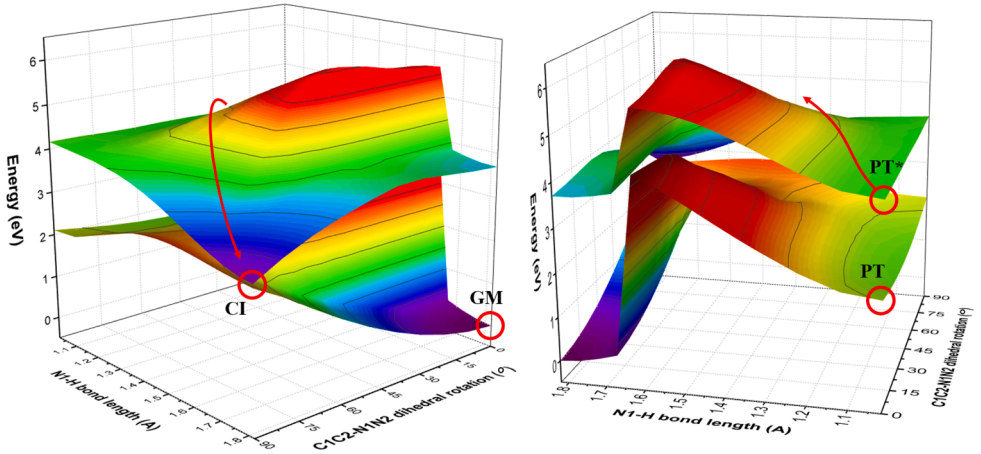


Fig. 8. PES around PT state and CI, colored by the relative energies (0–4.5 eV for S_0 surface, 2.5–7 eV for S_1 surface). Front view and back view.

Table 1
The calculated rate constants of PT*, TICT and TICT' states.

	$\Delta E^{\neq}$ (eV)	k^{TST} (s^{-1})	ω (eV)	f	k_r (s^{-1})
PT*	0.214	1.59×10^{9}	2.726	0.243	7.83×10^{7}
TICT	0.137	3.12×10^{10}	2.875	0.034	1.22×10^{7}
TICT'	0.134	3.51×10^{10}	2.889	0.044	1.59×10^{7}

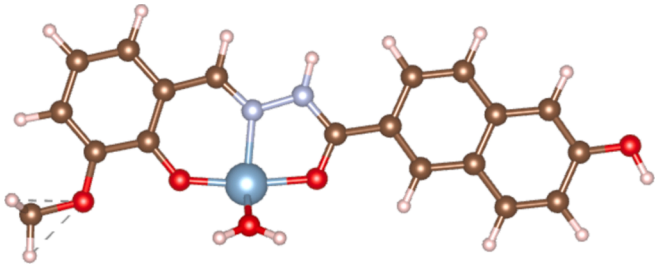


Fig. 9. The structure of 5-coordinated Al^{3+}-VN62 complex. The C, N, O, H, Al atoms are represented by brown, silver, red, white and light-steel blue, respectively (visualized in VESTA).

simultaneously, we will have the PES around the PT state and the CI. The variations of S_0/S_1 energies are plotted in Fig. 8. The GM, PT, PT* states and the CI can also be found in the PES. Similarly, there is a pathway between the PT* state and the CI in S_1 surface (red arrow in Fig. 8). The transition state (TS*-PT) and the energy diagram are also plotted in Figure S8.

The transition rate can be estimated by the energy barrier through TS*. Based on the transition state theory, [44] the equation of rate is written as:

$$k^{TST} = \frac{k_B T}{h} Q e^{-\frac{\Delta G^{\neq}}{k_B T}}$$

k_B is the Boltzmann constant, h is the Plank's constant, T is the temperature, Q is the quantum factor, which can be set as 1 in our calculation, and $\Delta G^{\neq}$ is free energy barrier. We use the electronic energy instead of the free energy, and the approximation is reasonable to calculate the quantity of the rate. The results are listed in Table 1.

The radiative rate of the fluorescent emission is calculated by using the emission energy ($\Delta E_{i,j}$) and the transition dipole moment ($\mu_{i,j}$): [27,45]

$$k_{rad} = \frac{4(\Delta E_{ij})^3 \mu_{ij}^2}{3\hbar^4 c^3}$$

The results are listed and compared with the transition rates of the CI pathways in Table 1. We can see that the rate of emission is much smaller than the rate of non-radiative transition, so that explains the fluorescent quenching of the probe.

3.5. Sensing mechanism for Al^{3+} detection

In summary, the de-excitation of the system is mostly achieved by the non-radiative CI pathways. Despite there are few stationary points in S_1 states (PT*, TICT and TICT'), these possible fluorescent states are quenched by transition to the non-radiative CI structures. So, the probe molecule alone has no fluorescence.

When the probe molecule interacts with the target Al^{3+}, a coordination complex will be formed, which induces the turn-on fluorescence. To confirm the interaction pattern between the metal ion and the ligand, the electrostatic potential is analyzed and plotted on molecular surface for the GM structure in ground state. As shown in Figure S9, the blue region corresponds to the negative value, and the red region corresponds to the positive value. Therefore, Al^{3+} can strongly coordinate to the negative charge cavity of the probe molecule. Combining with the experimental results [32] (including the Jobs plot, the ESI-MS spectra and ^{1}H NMR data), a 5-coordinated structure of the complex is deduced. Then the geometry of the Al^{3+}-VN62 complex is fully optimized and displayed in Fig. 9. The binding energy is calculated by the reaction: $Al^{3+} + L + 3H_2O = [Al(H_2O)_2L]^{2+} + H_3O^+$, and the corresponding $\Delta E = -464.1$ kcal/mol, which is consistent with previous data [12]. This binding energy is very strong, so the rotation of C—N bond in organic ligand will no longer happen.

Next, TDDFT calculation based on the above ground state structure and the optimization of S_1 state have been done to get the absorption and fluorescence spectra of the complex. The excitation information of complex $[Al(H_2O)_2L]^{2+}$ together with anion form of ligand L^- are listed

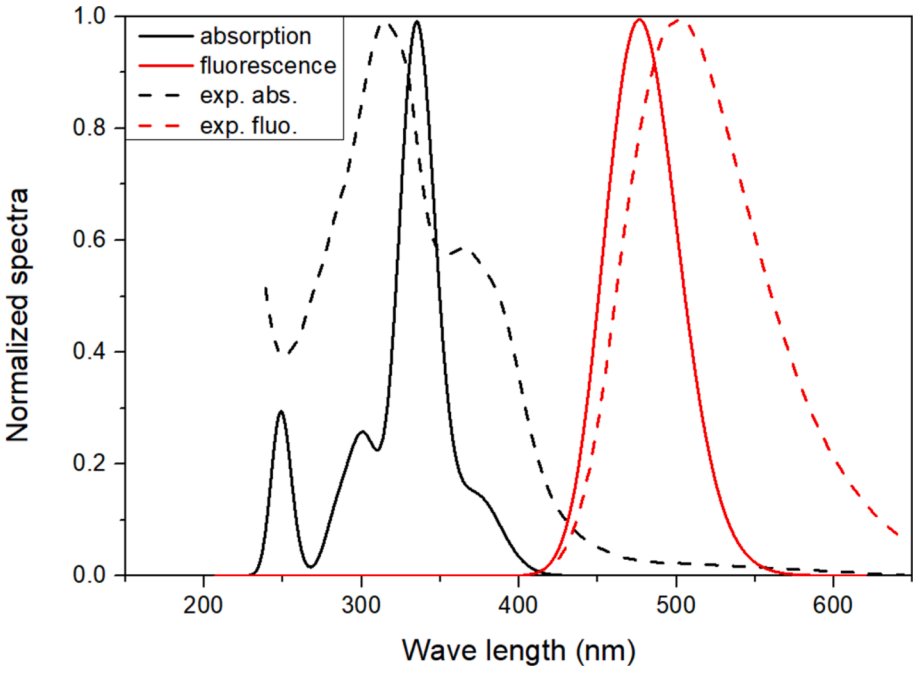


Fig. 10. Normalized absorption and fluorescence spectra of the complex in water.

in Table S2. The normalized absorption and fluorescence spectra of the complex comparing with the experimental spectra are plotted in Fig. 10. The highest absorption peak is contributed by the excitation of S_2 state, which is in 335 nm (3.70 eV), and the excitation of S_1 state has a small oscillator strength, shown a shoulder in 375 nm (3.31 eV). The fluorescence peak is in 476 nm (2.60 eV), giving a quite large Stokes shift of 0.71 eV. We can see that the peak positions of calculated spectra match well with the experimental data.

In the end, the excitation analysis is performed for the complex $[Al(H_2O)_2L]^{2+}$, and the electron-hole distribution of the complex in S_1 state is plotted in Figure S10. The excitation is mostly contributed by the ligand alone, and the similar excitation character is also found in the anion ligand L^-. Therefore, after coordination with Al^{3+}, the complex becomes emissive, and the fluorescence is mainly determined by the ligand part.

4. Conclusions

We conducted a theoretical investigation into the sensing mechanism of the acylhydrazone fluorescent probe VN62 for the detection of Al^{3+} ion. In the absence of the Al^{3+}, the molecule is non-fluorescent due to the non-radiative process involving conical intersections. The fluorescent S_1 states (PT*, TICT, TICI') can be formed after directly excitation from ground states or through internal conversion or intersystem crossing from higher excited states. However, these states are not stable enough to emit light, as they readily transition to the conical intersection. However, upon coordination with Al^{3+}, the CI process is suppressed, leading to the observation of fluorescence. This study aims to enhance our understanding of fluorescent probe sensing mechanisms at the molecular level and provide insights for designing future turn-on probes based on acylhydrazone derivatives.

CRediT authorship contribution statement

Pengcheng Zhang: Writing – original draft, Visualization, Software, Methodology, Investigation. **Chunxiang Zhao:** Validation, Conceptualization. **Hongyu Chen:** Funding acquisition, Formal analysis. **Han Zhang:** Writing – review & editing. **Huijie Ni:** Writing – review & editing. **Yubing Si:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.comptc.2024.114934.

Data availability

Data will be made available on request.

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